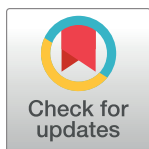


RESEARCH ARTICLE

Reward-related self-agency is disturbed in depression and anxiety

Marishka M. Mehta^{1,2}, Soojung Na³, Xiaosi Gu³, James W. Murrough^{1,2}, Laurel S. Morris^{1,2*}

1 Department of Psychiatry, Icahn School of Medicine at Mount Sinai, New York, NY, United States of America, **2** Depression and Anxiety Center for Discovery and Treatment, Icahn School of Medicine at Mount Sinai, New York, NY, United States of America, **3** Center for Computational Psychiatry, Icahn School of Medicine at Mount Sinai, New York, NY, United States of America

* laurel.morris@mssm.edu

Abstract

Background

The sense of agency, or the belief in action causality, is an elusive construct that impacts day-to-day experience and decision-making. Despite its relevance in a range of neuropsychiatric disorders, it is widely under-studied and remains difficult to measure objectively in patient populations. We developed and tested a novel cognitive measure of reward-dependent agency perception in an in-person and online cohort.

Methods

The in-person cohort consisted of 52 healthy control subjects and 20 subjects with depression and anxiety disorders (DA), including major depressive disorder and generalized anxiety disorder. The online sample consisted of 254 participants. The task consisted of an effort implementation for monetary rewards with computerized visual feedback interference and trial-by-trial ratings of self versus other agency.

Results

All subjects across both cohorts demonstrated higher self-agency after receiving positive-win feedback, compared to negative-loss feedback when the level of computer inference was kept constant. Patients with DA showed reduced positive feedback-dependent agency compared to healthy controls. Finally, in our online sample, we found that higher self-agency following negative-loss feedback was associated with worse anhedonia symptoms.

Conclusion

Together this work suggests how positive and negative environmental information impacts the sense of self-agency in healthy subjects, and how it is perturbed in patients with depression and anxiety.

OPEN ACCESS

Citation: Mehta MM, Na S, Gu X, Murrough JW, Morris LS (2023) Reward-related self-agency is disturbed in depression and anxiety. PLoS ONE 18(3): e0282727. <https://doi.org/10.1371/journal.pone.0282727>

Editor: Haijiang Li, Shanghai Normal University, CHINA

Received: July 13, 2022

Accepted: February 21, 2023

Published: March 15, 2023

Copyright: © 2023 Mehta et al. This is an open access article distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Data Availability Statement: The datasets generated during and/or analyzed during the current study are available. The associated de-identified data set is available at <https://github.com/laurelmorris/data>.

Funding: Funding was provided by NIMH/K01MH120433 for LSM. Additional funding was provided by the Ehrenkranz Laboratory for Human Resilience and the Friedman Brain Institute, both components of the Icahn School of Medicine at Mount Sinai. XG is supported by National Institute of Mental Health [grant number: R21MH120789,

R01MH122611, R01MH123069]. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Competing interests: The authors have no conflicts of interest to declare.

Introduction

The concept of self has long been debated in the field of psychology [1–5] due to its central role in manifestation of behavior. The study of self-processing and self as an object allows for characterizing self-relevant distortion in the context of psychopathology. Studies have suggested that when presented with a self-relevant stimulus, participants are biased towards themselves [6–9]. The nature of self is often studied as embodied (or bodily self) [10, 11], integrative self [9, 12–14], as the self vs other related valuations [15] or in social context [16] in empirical settings. A complementary concept to this is the sense of agency, belief of action ownership, or belief in a direct relationship between action and outcome, which can influence decision-making and motivated behavior.

A higher sense of self-agency, also known as internal locus of control, describes a stronger link between internal self-agency and environmental outcomes, whereas a lower sense of self-agency, known also as external locus of control, describes outcomes that are not tied to ones' actions. The sense of self-agency is a relatively stable trait (over 9 years), increasing slightly with age [17]. Higher self-agency has been associated with resilience to stress in the form of lower cortisol response to stress induction [18], greater incentive motivation [19], greater exploratory behavior [20], more flexible and goal-directed behaviors [19] and overall well-being [21]. Higher self-agency is also associated with higher plasma dopamine metabolite levels [22], and larger volumes of dopamine-rich fronto-striatal regions associated with cognitive and emotional control (i.e. ACC, striatum, anterior insula [23]). On the other hand, lower self-agency has been associated with negative emotionality [24], learned helplessness [25], higher risk of negative symptoms related to apathy and anhedonia [26] with early life experiences of low self-agency creating higher risk for later-life problematic anxiety [27].

Given these relationships between self-agency and prominent behavioral disturbances known to be disturbed in depression and anxiety disorders, it is surprising that little empirical evidence measures this trait in patient groups. Limited evidence mainly from self-reported measures of self-agency in the form of general locus of control (internal versus external) points towards a role for disrupted self-agency in patients with anxiety disorders and depression. One study in patients with major depressive disorder (MDD) showed an inverse relationship between self-agency (Rotter's LOC scale) and self-reported depressive symptoms (BDI) [28]. Another study in a large longitudinal cohort demonstrated that lower self-agency scores (CNSIE scale) at age 16 were associated with worse depressive symptoms at age 18, the former driven by lower early socioeconomic status [29]. Another longitudinal study in a large sample of subjects with a depressive or anxiety disorder found lower self-agency was associated with worse depression (Inventory of depression) and anxiety (BAI), with higher depression and negative life events predicting lower self-agency [17].

In this study, we propose a novel objective cognitive assessment of self-agency. We developed and operationalized an objective non-learning measure of self-agency in changing positive and negative environmental contexts and outcomes, to examine putative trait-level constructs of reward-dependent self-agency across groups. This task was tested in two samples, in the context of a laboratory-based in-person experiment as well as a larger online experiment. We predicted that patients with depressive and anxiety disorders, or those with worse depressive symptoms, would have overall lower self-agency across both positive and negative outcomes.

Methods

Experiment 1: In person participants

A community sample of volunteers between the ages of 18–65 were recruited through the Depression and Anxiety Center at the Icahn School of Medicine at Mount Sinai, New York.

Subjects were included if they met criteria for MDD or GAD as their primary psychiatric diagnosis as determined by the Structured Clinical Interview for DSM-V Axis Disorders (SCID-V) conducted by a trained rater. Subjects with MDD or GAD, were allowed comorbid mood or anxiety disorders, because of its prevalence [30]. Healthy subjects were free from any current or lifetime psychiatric disorder. For all groups, subjects were excluded if they had an unstable medical illness, history of neurological disease, neurodevelopmental or neurocognitive disorder, or positive urine toxicology test. Subjects completed cognitive testing and self-reported scales on the same day; this included the Temporal Experience of Pleasure Scale (TEPS), which is designed to capture dimensional measure of anhedonia. All methods were performed in accordance with the relevant guidelines and regulations set by the Program for Protection of Human Subjects (PPHS)/Institutional Review Board (IRB) at Icahn School of Medicine at Mount Sinai—approved written informed consent was obtained and subjects were compensated for their time.

Experiment 2: Online participants

A sample of US-based volunteers between the ages of 18–45 were recruited through the Prolific web-based research platform (<https://www.prolific.co/>). A total of 318 participants were enrolled in the study. Subjects completed an online version of the cognitive task and self-reported scales in the same session. Attention checks were embedded throughout the self-report questionnaires and any subjects that failed more than one attention check were excluded. Participants were also excluded if cognitive task performance-based data quality thresholds were not met. Performance-based data quality thresholds included performance of at least 10 button presses on average throughout the entire task and an overall average minimum reaction time of 1s, to exclude automatic responses without task engagement. The testing and data collection protocol was approved by the Program for Protection of Human Subjects (PPHS)/Institutional Review Board (IRB) at Icahn School of Medicine at Mount Sinai and subjects were compensated for their time.

The self-agency task (SAT)

The self-agency task was a 20-minute cognitive task programmed in *PsychoPy*. Throughout the task, subjects performed a simple effort task for reward over 108 trials (Fig 1). Subjects were instructed to press the left and right keyboard buttons quickly to move a visual bar on the computer screen to a target position. During baseline practice trials, reaction times were recorded and used to determine the subsequent threshold ($\times 1.25$). Each trial indicated the target bar position and context information on whether the trial was a win trial (for \$0.50) or a loss trial (-\$0.50). Subjects then performed the button presses to move the bar up to the target position as quickly as possible on each trial. Subjects received outcome feedback based on their performance (win trials: win money or nothing; loss trials: lose money or nothing). Performance was defined as successfully moving the bar to the target position in the available time, i.e. 125% of their average baseline reaction time.

Subjects were instructed that during the task there may be computer interference as they perform the simple effort task. During the task, there were 3 conditions of computer interference to modulate experience of agency, which was not known to the subject. The *Self* condition includes 18 trials with no computer interference, whereby each pair of button presses (i.e. left + right) moved the bar by 1 position, thus 4 button presses moves the bar up by 2 positions, etc. The *Computer* condition includes 18 trials with maximum computer interference whereby button presses did not directly equate to incremental bar movement. Here, for each bar position, a randomly determined number of button presses (between 1–6) was required to move

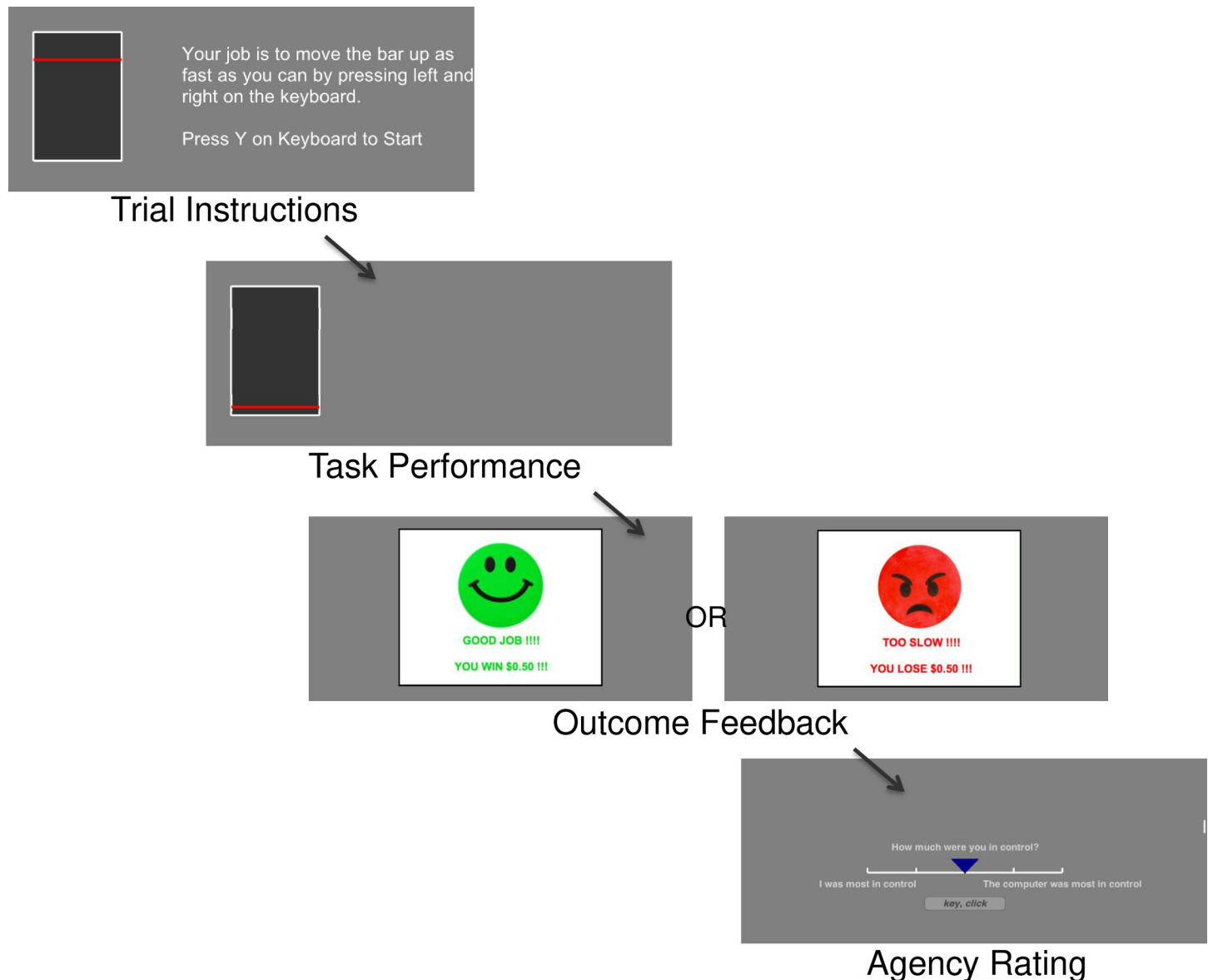


Fig 1. Self-agency task (SAT) schematic. On each trial, subjects perform a simple button press effort task for monetary reward. Subjects then gain positive-win or negative-loss feedback based on their reaction time relative to their own baseline reaction time. Following feedback, subjects are instructed to rate the level of agency for that trial on a 5-point Likert scale (1 = self, 5 = computer).

<https://doi.org/10.1371/journal.pone.0282727.g001>

the bar by 1 position such that there was a dissociation between button presses and visual feedback of bar movement. The *Ambiguous* condition included 72 trials with a mix of trials that had mild computer interference (between 1–3 button presses required to move the bar by 1 position). At the end of each trial, subjects rated who was more “in control” during that trial on a 5-point Likert scale where 1 = self, 5 = computer (Fig 1). The majority of trials were *Ambiguous* as performance during this condition is the main behavioral measure of interest. The *Self* and *Computer* condition trials were included firstly as contrast trials to objectively indicate the varying levels of agency and secondly to allow measurement of overall task accuracy and understanding. Subjects undertook training on the task in a self-paced manner before starting the task, including instruction and experience of the maximum computer interference

i.e., the “computer” condition. Subjects received bonus monetary payment depending on performance.

The online version of the SAT task was adapted for online testing and was identical to the in-person version. The task was created with *PsychoPy3* and hosted on their online platform, Pavlovia (<https://pavlovia.org/>).

Statistics

In order to determine the impact of outcome on self-agency, agency ratings following positive-win outcomes and negative-loss outcomes (when computer interference was kept constant in *Ambiguous* trials) were subjected to paired-samples t-tests for all subjects. Second, to examine group differences in self-agency, a two-tailed independent samples t-test was conducted to determine differences in self-agency between HC and DA. Finally, a mixed-effects linear regression was conducted to identify how all variables predicted self-agency ratings, including the effects of outcome feedback (i.e., win/loss), condition (i.e., self/computer/ambiguous), and group (i.e., HC/DA) on agency rating (rating $\sim 1 + \text{group} + \text{agency} + \text{feedback} + \text{group:agency} + \text{group:feedback} + (1 + \text{agency} + \text{feedback} \mid \text{subject})$). A separate mixed-effects linear regression was also conducted using separate groups of HC, MDD (primary) and GAD (primary) on an exploratory basis. The regression analysis was repeated with the online sample to evaluate how feedback and agency predicted ratings (rating $\sim \text{agency} + \text{feedback} + (1 + \text{agency} \mid \text{subject}) + (1 + \text{feedback} \mid \text{subject})$). A mixed-effects linear model was used to account for individual differences. The regressions were carried out using the *fitlme* function in Matlab.

Results

Experiment 1: In person participants

A total of 52 HC subjects performed the task. Two HC subjects did not fully complete the task, leaving 50 HC subjects for analysis (age = 39.4F0B19.9, 21 female). Twenty patients with DA disorders (age = 33.4F0B18.8, 13 female) completed the task including 7 MDD primary, 13 GAD primary, with the majority of DA subjects experiencing significant comorbidity (see [Table 1](#) for demographics).

Self-agency task. All subjects demonstrated higher self-agency after receiving positive-win feedback, compared to negative-loss feedback during the *Ambiguous* trials when the level

Table 1. Participant characteristics for Experiment 1.

	Healthy Controls	Depression and Anxiety
N	50	20
Age (mean $\pm$ SD)	39.4 $\pm$ 9.93	33.35 $\pm$ 8.89
Females (frequency, %)	21, 42%	15, 75%
Hispanic ethnicity (frequency, %)	7, 14%	4, 20%
Employed, at least part-time (frequency, %)	30, 60%	7, 35%
Some college (frequency, %)	22, 90%	17, 85%
Married (frequency, %)	13, 26%	4, 20%
N	32	17
Anhedonia (anticipatory)	41.75 $\pm$ 8.04	40.18 $\pm$ 7.67
Anhedonia (consummatory)	37.34 $\pm$ 6.99	38.18 $\pm$ 8.88

Participant characteristics for patients with depression and anxiety (DA) and healthy controls (HC). Anhedonia is measured with the temporal experience of pleasure scale (TEPS).

<https://doi.org/10.1371/journal.pone.0282727.t001>

of computer inference was kept constant (Fig 2, $T(69) = -6.25$, $p = 1.5 \times 10^{-8}$). This remained the case for the HC group ($T(49) = -5.04$, $p = 0.7 \times 10^{-5}$) and the DA group ($T(19) = -4.07$, $p = 0.001$), separately (Fig 2).

Higher self-agency following the positive-win feedback was associated with higher age across all subjects ($R = -0.308$, $p = 0.006$) and for the HC group ($R = -0.314$, $p = 0.045$) and the DA group ($R = -0.470$, $p = 0.037$), separately (Fig 3). There was no relationship between age and self-agency following negative-loss feedback ($p's > 0.3$), indicating a specificity of this bias in the positive domain. There were no differences within any groups or across groups based on sex ($p's > 0.5$).

Patients with DA disorders reported lower self-agency following positive-win feedback compared to healthy controls ($T(48) = -2.171$, $p = 0.035$) (Fig 4), with no difference following negative-loss feedback ($T(68) = -1.475$, $p = 0.145$).

Mixed-effects linear regression model. A mixed-effects linear regression was conducted to examine all effects, including group (HC/DA), agency condition (*Self*, *Computer*, *Ambiguous*), and outcome feedback (positive-win, negative-loss) on agency rating.

First, there was a significant effect of agency condition whereby self-agency was higher in the *Self* group (the reference group in the regression) than the *Ambiguous* condition ($\beta = 0.40$,

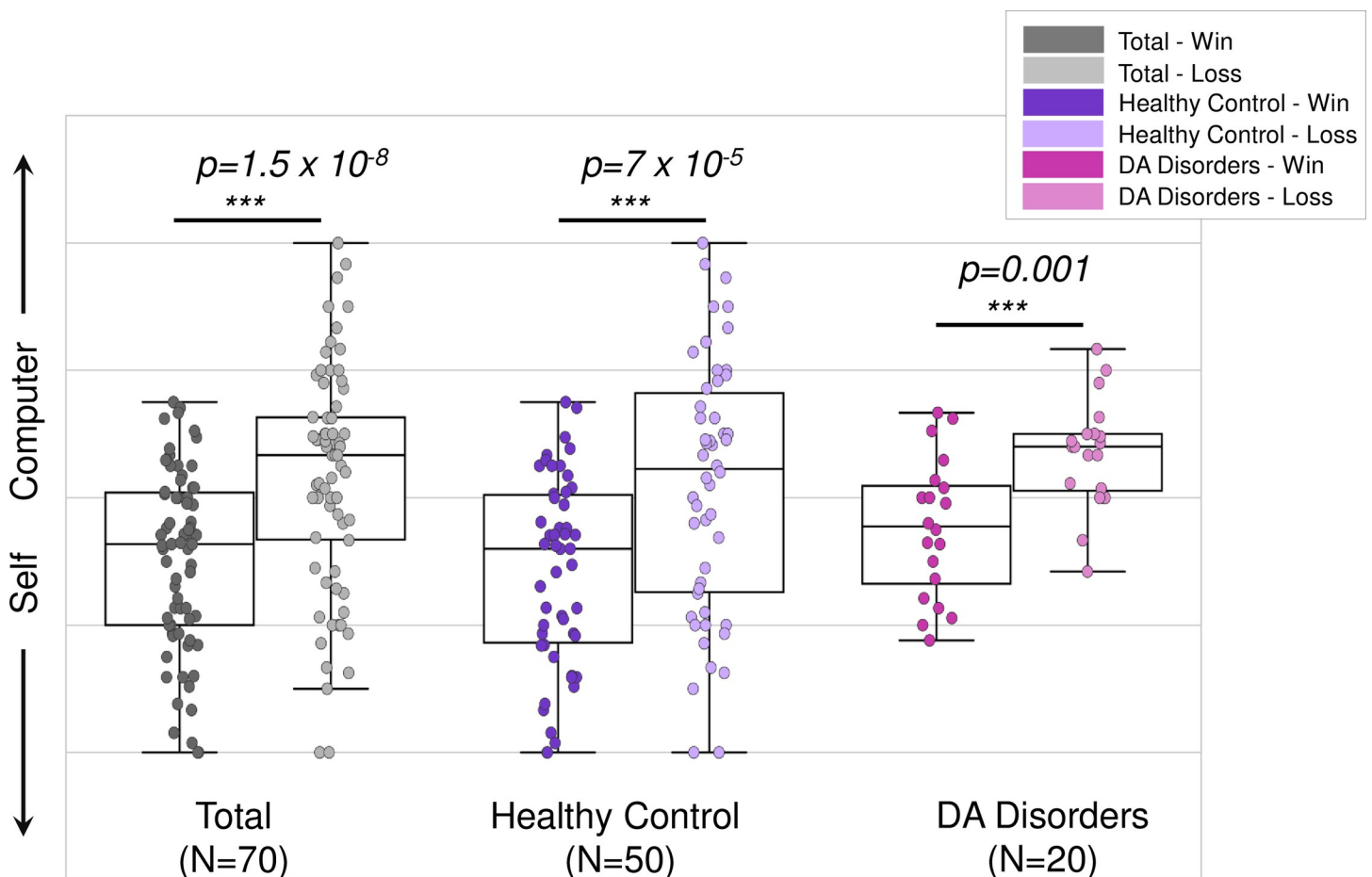


Fig 2. Higher positive feedback-related self-agency across all groups. Agency ratings plotted following positive-win (darker colors) and negative-loss (lighter colors) feedback during Ambiguous trials for all subjects ($N = 70$), healthy control subjects only ($N = 50$) and subjects with depression and anxiety (DA) disorders ($N = 20$). All subjects demonstrated higher self-agency following positive-win outcomes compared to negative-loss outcomes.

<https://doi.org/10.1371/journal.pone.0282727.g002>

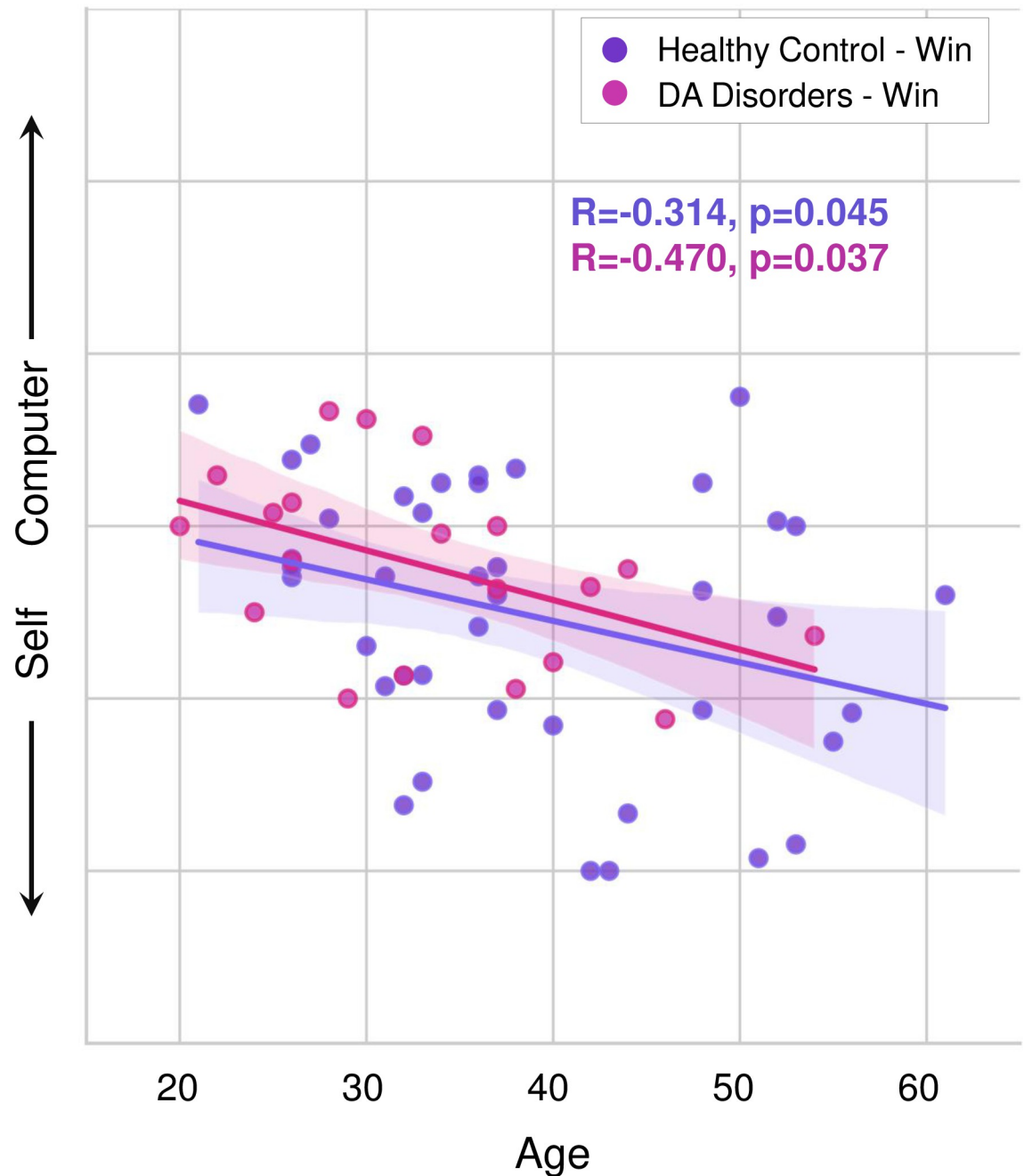


Fig 3. Higher positive feedback-dependent self-agency increases with age across all groups. Agency ratings plotted following positive-win feedback against age in healthy control subjects ($N = 50$, purple) and subjects with depression and anxiety (DA) disorders ($N = 20$, pink).

<https://doi.org/10.1371/journal.pone.0282727.g003>

$p < 0.001$) or the *Computer* condition ($\beta = 1.16$, $p < 0.001$), meaning subjects understood the task and performed accurately. Second, there was a significant effect of feedback ($\beta = -0.66$, $p < 0.001$) whereby there was higher self-agency following positive-win compared to negative-loss feedback, corroborating the primary test above. Third, there was no significant main effect of group ($\beta = 0.01$, $p = 0.97$), but there was a significant interaction between group and

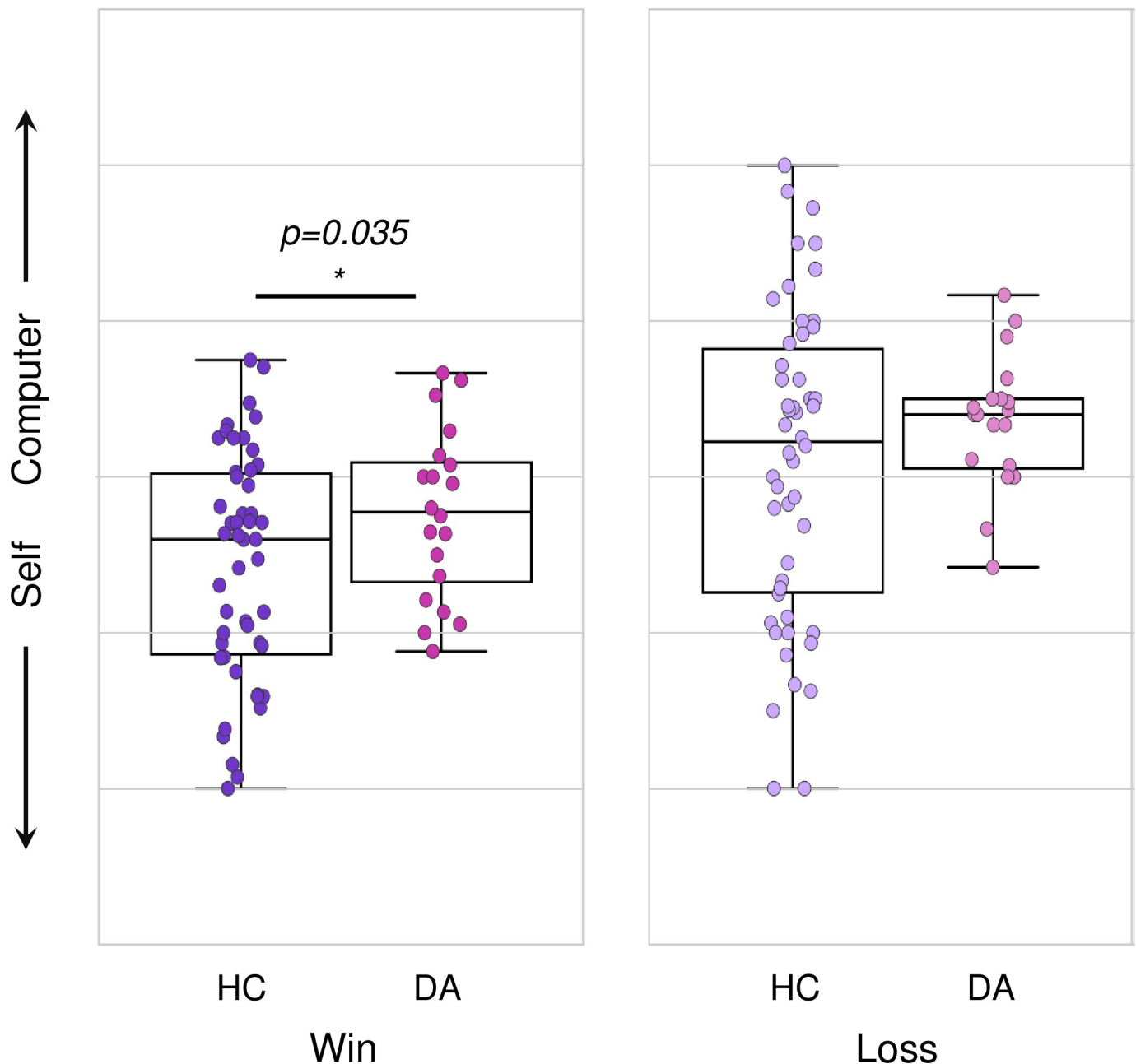


Fig 4. Lower negative feedback-dependent self-agency in depression and anxiety disorders. Agency ratings plotted following positive-win and negative-loss feedback during Ambiguous trials, plotted separately for healthy control subjects ($N = 50$) and subjects with depression and anxiety (DA) disorders ($N = 20$) (data is recapitulated from Fig 1).

<https://doi.org/10.1371/journal.pone.0282727.g004>

feedback ($\beta = 0.27$, $p < 0.05$), whereby DA subjects had lower self-agency following positive-win feedback compared to the HC group.

The second exploratory regression model that included separate denotations of MDD-primary and GAD-primary groupings produced largely similar results and suggested that both MDD-primary subjects ($\beta = 0.38$, $p < 0.05$) and GAD-primary subjects ($\beta = 0.23$, $p = 0.06$) had lower self-agency following positive-win feedback compared to HC. See S1 and S2 Tables for all results from the mixed-effects regression models.

Dimensional symptom domains. Relationships between self-agency and anhedonia (TEPS) were examined in an exploratory manner. There were no significant relationships between self-agency and dimensional measures of anhedonia in HC and DA groups.

Experiment 2: Online participants

A total of 254 participants performed the task, of which, one participant did not fully complete the task. After the preliminary quality check, seven participants were excluded, leaving a sample size of 246 for analysis (age = 32.83 F0B16.54, 127 female). See Table 2 for subject characteristics.

Self-agency task. Similar to the in-person findings, all subjects demonstrated higher self-agency after receiving positive-win feedback, compared to negative-loss feedback during the *Ambiguous* trials when the level of computer inference was kept constant (Fig 5A, $T(181) = 15.07$, $p = 8.57 \times 10^{-34}$).

Higher self-agency following the positive-win feedback showed a negative relationship with higher age across all subjects, which did not reach significance ($R = -0.1$, $p = 0.135$). The correlation was likely not significant due to the restricted age range of this sample (ages 18–45). Again, there was no relationship between age and self-agency following negative-lose feedback, and no differences based on sex (p 's > 0.5).

Mixed-effects linear regression model. A mixed-effects linear regression was conducted to examine all effects, including agency condition (Self, Computer, Ambiguous), and outcome feedback (positive-win, negative-loss) on agency rating. First, we replicated the significant effect of agency condition ($\beta = 0.39$, $p = 6.13 \times 10^{-54}$) whereby self-agency was higher in the Self group (the reference group in the regression) than the Ambiguous condition ($T(490) = -6.59$, $p = 1.16 \times 10^{-10}$) and the Computer condition ($T(490) = -11.28$, $p = 2.24 \times 10^{-26}$), meaning subjects understood the task and performed accurately.

Second, there was also a significant effect of feedback ($\beta = 1.10$, $p = 1.78 \times 10^{-64}$) whereby there was higher self-agency following positive-win compared to negative-loss feedback ($T(343) = -14.05$, $p = 9.82 \times 10^{-36}$), corroborating the primary test above. See S3 Table for the results from the mixed-effects regression model.

Dimensional symptom domains. Relationships between self-agency and anhedonia were again examined in an exploratory manner across all online subjects. Contrary to the findings in Experiment 1, higher self-agency following negative-loss feedback was associated with

Table 2. Participant characteristics for Experiment 2.

Online sample characteristics	
N	246
Age (mean $\pm$ SD)	32.83 $\pm$ 6.54
Females (frequency, %)	127, 51.62%
Hispanic ethnicity (frequency, %)	19, 7.720%
Employed, at least part-time (frequency, %)	190, 77.23%
Some college (frequency, %)	212, 86.18%
Married (frequency, %)	107, 43.5%
Anhedonia (anticipatory)	42.49 $\pm$ 7.86
Anhedonia (consummatory)	37.09 $\pm$ 6.19

Participant characteristics for participants from the online study. Anhedonia is measured with the Temporal experience of pleasure scale (TEPS).

<https://doi.org/10.1371/journal.pone.0282727.t002>

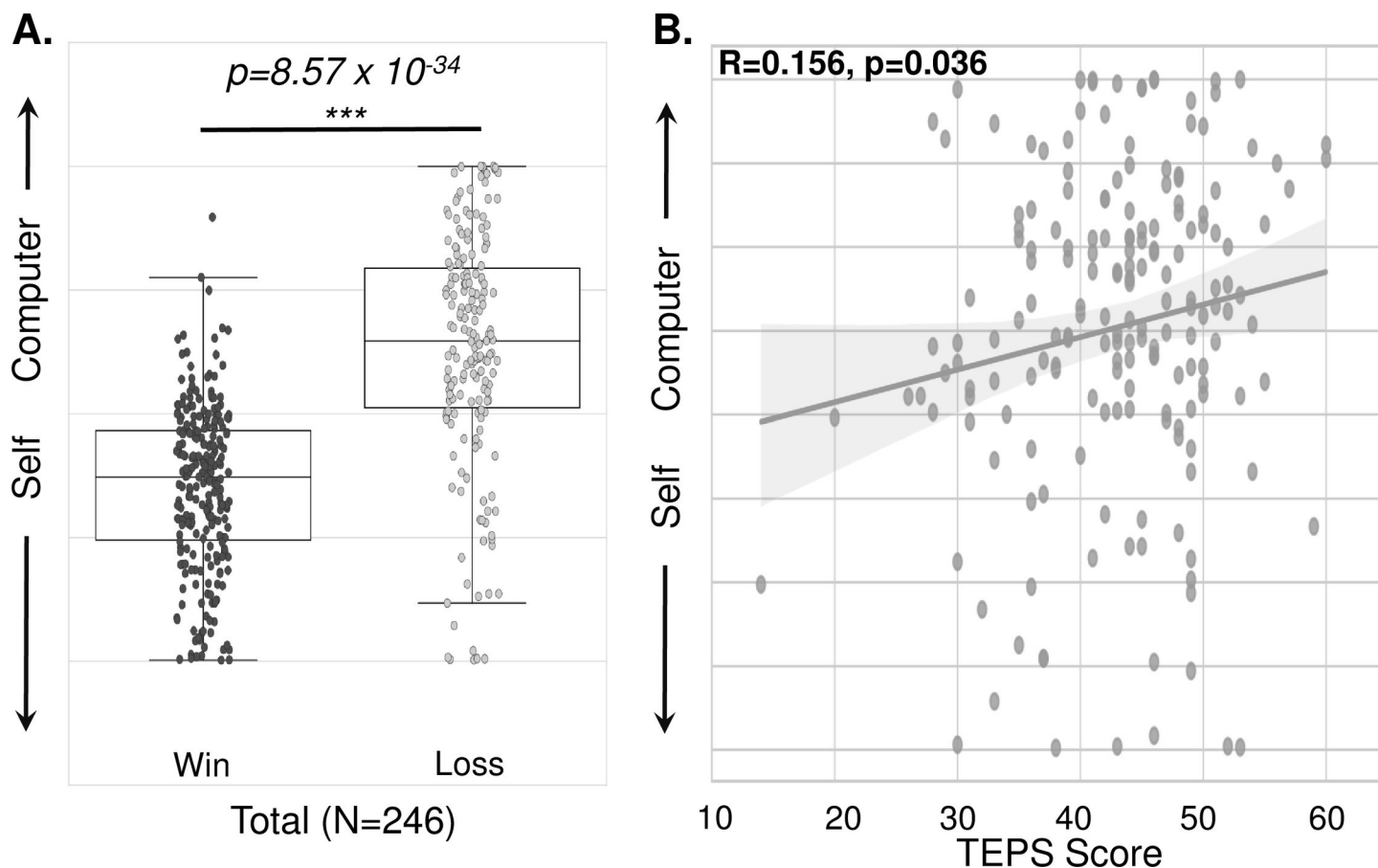


Fig 5. Higher positive feedback-dependent agency replicated in online sample. A, Agency ratings plotted following positive-win and negative-loss feedback during Ambiguous trials for all subjects in the online session ($N = 246$). B, Agency ratings plotted following negative-loss feedback during Ambiguous trials against Temporal Experience of Pleasure Scale (TEPS) anticipatory sub-score for the online sample ($N = 246$). Higher TEPS scores represent higher pleasure ratings, thus being inversely related to anhedonia.

<https://doi.org/10.1371/journal.pone.0282727.g005>

worse symptoms of anhedonia (TEPS-anticipatory) in the online sample ($R = 0.156$, $p = 0.036$, Fig 5B).

Discussion

This paper showcases an objective study of self-agency in a dynamic environment in healthy individuals and subjects with mood and anxiety disorders. Since self-agency has been associated with syndromes related to anhedonia in depression [24–26] and high risk for problematic anxiety [27], this study focused on the symptom domain of anhedonia and its relationship with sense of self-agency in the context of positive and negative feedbacks. The task was also repeated in an online sample, representative of a general population, to capture variations in self-reported anhedonia and relationships with agency.

Across all subjects in both experiments, there was higher self-agency in relation to positive-win feedback and lower self-agency in relation to negative-loss feedback. This positive-win feedback-dependent agency generally increased with increasing age. Subjects with mood and anxiety disorders showed lower negative feedback-dependent agency compared to healthy controls in the win trials, with no difference during the loss trials. Higher self-agency in response to negative-loss feedback was also associated with worse self-reported anhedonia

symptoms in our online participants. Together, these results indicate a novel belief bias that presents across all subjects and suggests a potential trans-diagnostic disturbance related to reduction in self-agency.

The current results corroborate literature on self-agency bias in healthy subjects that increases with age [17]. However, this extends upon previous work by utilizing an objective cognitive task that directly separates self-agency in response to positive feedbacks compared to negative feedbacks, highlighting that only the positive feedback-dependent agency increases with age.

Reduced self-agency reporting in response to negative feedback in subjects with DA disorders was expected given previous literature linking self-reported self-agency and self-reported depression [28] and anxiety [17] symptoms. These results were also replicated in our online sample. However, the current findings suggest that in this sample of patients, reduced self-bias or a more external locus of control exists more so in a positive environment only, rather than in a negative environment. This is surprising given a wealth of literature highlighting negative biases in depression and anxiety disorders. Selective memory recall for negative information [17] and negative bias related to perception of facial expressions [31] is higher in depressed cohorts. Similarly, attentional bias towards negative information [32] and negative interpretation bias of ambiguous information [33] is well established in anxiety disorders. However, while negative attentional bias is inconsistently reported in depression, this bias seems to increase for self-relevant negative information [34, 35], whereby depressed subjects relate negative information to self more than control subjects. This suggests that negative self-bias in depression may relate to negative self-evaluation, rather than attributing negative environmental outcomes to ones' actions. While there were no group differences in negative self-bias in these samples, higher negative self-bias was associated with anhedonia in a larger population-based cohort. A recent study also found that arousal levels might selectively modulate the self-relevant processes in depressive mood [36]. This role of arousal as a possible confound may also explain differences in the results.

Together, these results suggest an interesting avenue for exploring possible relationships between anhedonia symptoms related to lower reward-dependent agency in response to negative outcomes. However, these findings must be replicated in another study before confirming specific links between anhedonia and reward-dependent self-agency in response to negative feedback.

Limitations and future directions

Finally, while mood and anxiety disorders and symptoms are highly comorbid [30, 37], further studies in larger sample sizes must disentangle the relative contributions of reward-dependent self-agency to depression, anhedonia, stress and anxiety-related symptoms across and within disorder groups. Moving away from categorical diagnoses of disorders, it will be important to dissect how specific symptom domains, such as rumination or threat-reactivity relate to underlying disturbances in self-agency across groups. If these findings are replicated, an important future avenue of research would be to examine how agency belief-updating can be modulated using self-agency related feedback or training, in order to normalize aberrant self-agency biases related to symptoms across patient groups.

Supporting information

S1 Table. Results of the linear mixed effects model to test agency and feedback as predictors of rating (sense of agency) for patients with depression and anxiety disorder (DA). (DOCX)

S2 Table. Results of the linear mixed effects model to test agency and feedback as predictors of rating (sense of agency) for patients with primary generalized anxiety disorder (GAD) and major depressive disorder (MDD), which compromised the depression and anxiety disorder (DA) group.

(DOCX)

S3 Table. Results of the linear mixed effects model to test agency (self, ambiguous and computer) and feedback (positive-win and negative-loss) as predictors of rating (sense of agency) for the online sample.

(DOCX)

Author Contributions

Formal analysis: Marishka M. Mehta, Soojung Na, Xiaosi Gu, Laurel S. Morris.

Funding acquisition: Xiaosi Gu, Laurel S. Morris.

Supervision: Laurel S. Morris.

Visualization: Marishka M. Mehta, Laurel S. Morris.

Writing – original draft: Marishka M. Mehta, Laurel S. Morris.

Writing – review & editing: Marishka M. Mehta, Xiaosi Gu, James W. Murrough, Laurel S. Morris.

References

1. Christoff K, Cosmelli D, Legrand D, Thompson E. Specifying the self for cognitive neuroscience. *Trends Cogn Sci*. 2011; 15: 104–112. <https://doi.org/10.1016/j.tics.2011.01.001> PMID: 21288760
2. Northoff G. Self and brain: what is self-related processing? *Trends Cogn Sci*. 2011; 15: 186–187. <https://doi.org/10.1016/j.tics.2011.03.001> PMID: 21458358
3. Sui J, Gu X. Self as Object: Emerging Trends in Self Research. *Trends Neurosci*. 2017; 40: 643–653. <https://doi.org/10.1016/j.tins.2017.09.002> PMID: 28988827
4. Woźniak M. “I” and “Me”: The self in the context of consciousness. *Front Psychol*. 2018; 9: 1656. <https://doi.org/10.3389/fpsyg.2018.01656> PMID: 30233474
5. Gallagher S. Philosophical conceptions of the self: implications for cognitive science. *Trends Cogn Sci*. 2000; 4: 14–21. [https://doi.org/10.1016/s1364-6613\(99\)01417-5](https://doi.org/10.1016/s1364-6613(99)01417-5) PMID: 10637618
6. Heatherton TF, Wyland CL, Macrae CN, Demos KE, Denny BT, Kelley WM. Medial prefrontal activity differentiates self from close others. *Soc Cogn Affect Neurosci*. 2006; 1: 18–25. <https://doi.org/10.1093/scan/nsl001> PMID: 18985097
7. Cunningham SJ, Turk DJ, Macdonald LM, Macrae CN. Yours or mine? Ownership and memory. *Consciousness and Cognition: An International Journal*. Cunningham, Sheila J.: School of Psychology, University of Aberdeen, William Guild Building, Aberdeen, United Kingdom, AB24 2UB, sheila.cunningham@abdn.ac.uk: Elsevier Science; 2008. pp. 312–318. <https://doi.org/10.1016/j.concog.2007.04.003>
8. Sui J, Humphreys GW. The Interaction between Self-Bias and Reward: Evidence for Common and Distinct Processes. *Q J Exp Psychol*. 2015; 68: 1952–1964. <https://doi.org/10.1080/17470218.2015.1023207> PMID: 25851057
9. Ma Y, Han S. Why we respond faster to the self than to others? An implicit positive association theory of self-advantage during implicit face recognition. *Journal of Experimental Psychology: Human Perception and Performance*. Han, Shihui: Department of Psychology, Peking University, 5 Yiheyuan Road, Beijing, China, 100871, shan@pku.edu.cn: American Psychological Association; 2010. pp. 619–633. <https://doi.org/10.1037/a0015797>
10. Botvinick M, Cohen J. Rubber hands ‘feel’ touch that eyes see. *Nature*. 1998; 391: 756. <https://doi.org/10.1038/35784> PMID: 9486643

11. Fang W, Li J, Qi G, Li S, Sigman M, Wang L. Statistical inference of body representation in the macaque brain. *Proc Natl Acad Sci*. 2019; 116: 20151–20157. <https://doi.org/10.1073/pnas.1902334116> PMID: [31481617](https://pubmed.ncbi.nlm.nih.gov/31481617/)
12. Sui J, He X, Humphreys G. Perceptual effects of social salience: evidence from self-prioritization effects on perceptual matching. *J Exp Psychol*. 2012; 38: 1105–1117. <https://doi.org/10.1037/a0029792> PMID: [22963229](https://pubmed.ncbi.nlm.nih.gov/22963229/)
13. Sui J, Rotshtein P, Humphreys GW. Coupling social attention to the self forms a network for personal significance. *Proc Natl Acad Sci*. 2013; 110: 7607–7612. <https://doi.org/10.1073/pnas.1221862110> PMID: [23610386](https://pubmed.ncbi.nlm.nih.gov/23610386/)
14. Macrae CN, Visokomogilski A, Golubickis M, Cunningham WA, Sahraie A. Self-relevance prioritizes access to visual awareness. *J Exp Psychol Hum Percept Perform*. 2017; 43: 438–443. <https://doi.org/10.1037/xhp0000361> PMID: [28240929](https://pubmed.ncbi.nlm.nih.gov/28240929/)
15. King-Casas B, Tomlin D, Anen C, Camerer CF, Quartz SR, Montague PR. Getting to Know You: Reputation and Trust in a Two-Person Economic Exchange. *Science* (80-). 2005; 308: 78–83. <https://doi.org/10.1126/science.1108062> PMID: [15802598](https://pubmed.ncbi.nlm.nih.gov/15802598/)
16. Wimmer H, Perner J. Beliefs about beliefs: Representation and constraining function of wrong beliefs in young children's understanding of deception. *Cognition*. 1983; 13: 103–128. [https://doi.org/10.1016/0010-0277\(83\)90004-5](https://doi.org/10.1016/0010-0277(83)90004-5) PMID: [6681741](https://pubmed.ncbi.nlm.nih.gov/6681741/)
17. Hovenkamp-Hermelink JHM, Jeronimus BF, van der Veen DC, Spinhoven P, Penninx BWJH, Schoevers RA, et al. Differential associations of locus of control with anxiety, depression and life-events: A five-wave, nine-year study to test stability and change. *J Affect Disord*. 2019; 253: 26–34. <https://doi.org/10.1016/j.jad.2019.04.005> PMID: [31009845](https://pubmed.ncbi.nlm.nih.gov/31009845/)
18. Bollini AM, Walker EF, Hamann S, Kestler L. The influence of perceived control and locus of control on the cortisol and subjective responses to stress. *Biol Psychol*. 2004; 67: 245–260. <https://doi.org/10.1016/j.biopsycho.2003.11.002> PMID: [15294384](https://pubmed.ncbi.nlm.nih.gov/15294384/)
19. Declerck CH, Boone C, De Brabander B. On feeling in control: A biological theory for individual differences in control perception. *Brain Cogn*. 2006; 62: 143–176. <https://doi.org/10.1016/j.bandc.2006.04.004> PMID: [16806623](https://pubmed.ncbi.nlm.nih.gov/16806623/)
20. Kayser AS, Mitchell JM, Weinstein D, Frank MJ. Dopamine, locus of control, and the exploration-exploitation tradeoff. *Neuropsychopharmacology*. 2015; 40: 454–462. <https://doi.org/10.1038/npp.2014.193> PMID: [25074639](https://pubmed.ncbi.nlm.nih.gov/25074639/)
21. DeNeve KM, Cooper H. The happy personality: a meta-analysis of 137 personality traits and subjective well-being. *Psychol Bull*. 1998; 124: 197–229. <https://doi.org/10.1037/0033-2909.124.2.197> PMID: [9747186](https://pubmed.ncbi.nlm.nih.gov/9747186/)
22. De Brabander B, Declerck CH. A possible role of central dopamine metabolism associated with individual differences in locus of control. *Pers Individ Dif*. 2004; 37: 735–750. <https://doi.org/10.1016/j.paid.2003.11.001>
23. Hashimoto T, Takeuchi H, Taki Y, Sekiguchi A, Nouchi R, Kotozaki Y, et al. Neuroanatomical correlates of the sense of control: Gray and white matter volumes associated with an internal locus of control. *Neuroimage*. 2015; 119: 146–151. <https://doi.org/10.1016/j.neuroimage.2015.06.061> PMID: [26123375](https://pubmed.ncbi.nlm.nih.gov/26123375/)
24. Yoshie M, Haggard P. Negative emotional outcomes attenuate sense of agency over voluntary actions. *Curr Biol*. 2013; 23: 2028–2032. <https://doi.org/10.1016/j.cub.2013.08.034> PMID: [24094850](https://pubmed.ncbi.nlm.nih.gov/24094850/)
25. Soral W, Kofta M, Bukowski M. Helplessness experience and intentional (un-)binding: Control deprivation disrupts the implicit sense of agency. *Journal of Experimental Psychology: General*. Soral, Wiktor: Faculty of Psychology, University of Warsaw, Stawki 5/7, Warsaw, Poland, 00–183, wiktors.soral@psych.uw.edu.pl: American Psychological Association; 2021. pp. 289–305. <https://doi.org/10.1037/xge0000791>
26. Thompson A, Papas A, Bartholomeusz C, Nelson B, Yung A. Externalized attributional bias in the Ultra High Risk (UHR) for psychosis population. *Psychiatry Res*. 2013; 206: 200–205. <https://doi.org/10.1016/j.psychres.2012.10.017> PMID: [23177592](https://pubmed.ncbi.nlm.nih.gov/23177592/)
27. Chorpita BF, Barlow DH. The development of anxiety: the role of control in the early environment. *Psychol Bull*. 1998; 124: 3–21. <https://doi.org/10.1037/0033-2909.124.1.3> PMID: [9670819](https://pubmed.ncbi.nlm.nih.gov/9670819/)
28. Abdolmanafi A, Besharat MA, Farahani H, Khodaii MR. The moderating role of locus of control on the relationship between anger and depression in patients with major depression disorder. *Procedia—Soc Behav Sci*. 2011; 30: 297–301. <https://doi.org/10.1016/j.sbspro.2011.10.059>
29. Culpin I, Stapinski L, Miles ÖB, Araya R, Joinson C. Exposure to socioeconomic adversity in early life and risk of depression at 18 years: The mediating role of locus of control. *J Affect Disord*. 2015; 183: 269–278. <https://doi.org/10.1016/j.jad.2015.05.030> PMID: [26047304](https://pubmed.ncbi.nlm.nih.gov/26047304/)

30. Brown TA, Campbell LA, Lehman CL, Grisham JR, Mancill RB. Current and lifetime comorbidity of the DSM-IV anxiety and mood disorders in a large clinical sample. *J Abnorm Psychol.* 2001; 110: 585–599. <https://doi.org/10.1037//0021-843x.110.4.585> PMID: 11727948
31. Raes F, Hermans D, Williams JMG. Negative bias in the perception of others' facial emotional expressions in major depression: The role of depressive rumination. *J Nerv Ment Dis.* 2006; 194: 796–799. <https://doi.org/10.1097/01.nmd.0000240187.80270.bb> PMID: 17041294
32. Mogg K, Bradley BP. Attentional bias in generalized anxiety disorder versus depressive disorder. *Cognit Ther Res.* 2005; 29: 29–45. <https://doi.org/10.1007/s10608-005-1646-y>
33. Huppert JD, Foa EB, Furr JM, Filip JC, Mathews A. Interpretation bias in social anxiety: A dimensional perspective. *Cognit Ther Res.* 2003; 27: 569–577. <https://doi.org/10.1023/A:1026359105456>
34. Segal Z V, Gemar M, Truchon C, Guirguis M, Horowitz LM. A priming methodology for studying self-representation in major depressive disorder. *Journal of Abnormal Psychology.* US: American Psychological Association; 1995. pp. 205–213. <https://doi.org/10.1037/0021-843x.104.1.205> PMID: 7897044
35. Mogg K, Bradbury K, Bradley B. Interpretation of ambiguous information in clinical depression. *Behav Res Ther.* 2006; 44: 1411–1419. <https://doi.org/10.1016/j.brat.2005.10.008> PMID: 16487479
36. Qian H, Wang Z, Li C, Gao X. Prioritised self-referential processing is modulated by emotional arousal. *Q J Exp Psychol.* 2019; 73: 688–697. <https://doi.org/10.1177/1747021819892158> PMID: 31747826
37. Brown JD, Dutton KA, Cook KE. From the top down: Self-esteem and self-evaluation. *Cogn Emot.* 2001; 15: 615–631. <https://doi.org/10.1080/02699930126063>